

Problem Statement No. 39

Predictive Maintenance of Industrial Machinery

Domain

Mechanical Engineering (Machine Learning Project)

The Challenge

Develop a predictive maintenance model for a fleet of industrial machines to anticipate failures before they occur. This project involves analyzing sensor data from machinery to identify patterns that precede a failure. The goal is to create a classification model that can predict the type of failure (e.g., tool wear, heat dissipation, power failure) based on real-time operational data. This will enable proactive maintenance, reducing downtime and operational costs.

Objective

Build a machine learning model that can:

- Analyze sensor data (air temperature, process temperature, rotational speed, torque, tool wear) from industrial machines.
- Classify the type of failure a machine is likely to face — No Failure, Tool Wear Failure, Heat Dissipation Failure, Power Failure, Overstrain Failure, or Random Failure.
- Enable proactive maintenance by predicting failures before they happen, reducing unplanned downtime and repair costs.
- Provide a deployable, testable model that gives real-time predictions on new machine data.

Dataset

Kaggle: Machine Predictive Maintenance Classification

<https://www.kaggle.com/datasets/shivamb/machine-predictive-maintenance-classification>

Technology

Use of IBM Cloud Lite services is mandatory.